

Workshop in Machine Learning and Data Analysis

PREDICTING PROBLEMATIC INTERNET USE IN CHILDREN

<https://www.kaggle.com/competitions/child-mind-institute-problematic-internet-use>

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THE CHALLENGE

The competition's goal is to predict the level of problematic internet usage exhibited by children and adolescents, based on their physical activity.

Current methods for measuring problematic internet use in children and adolescents are often complex and require professional evaluation. Conversely, physical and fitness measures are extremely accessible and widely available with minimal intervention or clinical expertise.

The hypothesis is that changes in physical habits, such as reduced physical activity, irregular sleep patterns, or increased sitting time - may serve as early indicators of problematic internet use.

DATA OVERVIEW

The dataset for this competition is drawn from the Healthy Brain Network clinical research sample, covering approximately 5,000 participants aged between 5 and 22. In addition, some participants were given an accelerometer to wear for up to 30 days continually while going about their regular daily lives. Therefore, only a subset of the participants includes detailed accelerometer (actigraphy) data, while others have only questionnaire-based and demographic information.

Two main data sources are used in this competition:

1. Physical activity data (wrist-worn accelerometer data, fitness assessments and questionnaires).

2. Internet usage behavior data.

The accelerometer data is provided in parquet files, while the remaining tabular data is provided in CSV format.

The tabular data comprises measurements from a variety of instruments, including: Demographics, Internet Use, Children's Global Assessment Scale, Physical Measures, Bio-electric Impedance Analysis, Physical Activity Questionnaire, Sleep Disturbance Scale, and the Parent-Child Internet Addiction Test.

The actigraphy files provide continuous motion data collected over multiple days for each participant, which can be used to derive measures of activity and sleep.

The competition is evaluated using the Quadratic Weighted Kappa (QWK), a metric that checks how close our predictions are to the true SII (Severity Impairment Index) scores, while giving partial credit when the prediction is close but not exact. The goal of this competition is to predict each participants' SII, a standard measure of problematic internet use, based on the available physical and behavioral data.

IDEAL OUTCOMES

In today's digital age, problematic internet use among children and adolescents is a growing concern. Better understanding this issue is crucial for addressing mental health problems such as depression and anxiety.

We will consider the project successful if we achieve the following:

1. Develop a model that predicts the Problematic Internet Use (SII) score or category significantly better than simple models.
2. Derive meaningful Insights - Identify key behavioral, demographic, and psychological factors that are most strongly associated with problematic internet use (for example: sleep quality, activity levels, or age).
3. Deliver a clean and well-documented pipeline that includes visualizations and explanations of the model's results, making findings understandable and interpretable even for non-experts.

EXPECTED COMPUTE AND STORAGE REQUIREMENTS

Data size is 6.73 GB, containing 1002 files of type parquet and csv.

The competition is a code competition.

We estimate the project will require moderate computational resources.

INITIAL IDEAS

1. Visualize and summarize the data to identify correlations between physical activity patterns (e.g., daily movement, sleep duration) and the SII.
2. Detect missing values, outliers, and potential biases in the data. Start with simple models (e.g., Linear Regression) to establish a baseline and understand feature importance.
3. Advance to more complex models such as Gradient Boosting models (e.g., XGBoost) or deep learning approaches.
4. Analyze which behavioral or physiological patterns most strongly predict problematic internet use.
5. Translate findings into actionable insights about early detection and prevention of digital overuse among youth.

OUR PROGRESS

The table below presents only the improvements that led to measurable performance gains. Additional unsuccessful attempts are discussed later.

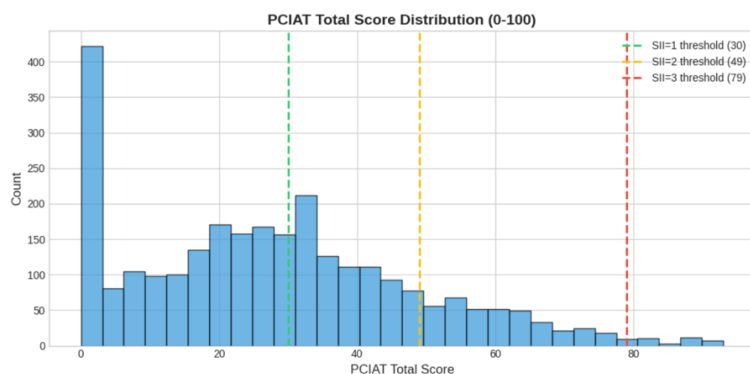
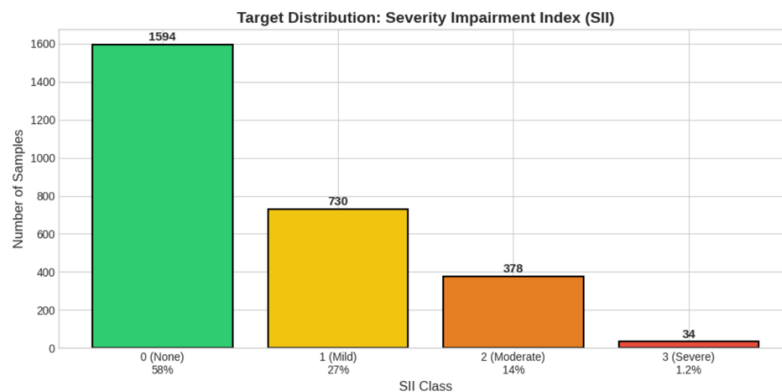
change	private score (QWK)
Baseline: single LGBM, median imputation, SII target	0.336
PCIAT_Total as target	0.419
threshold optimization	0.441
3-model ensemble (+ XGBoost, CatBoost) + KNN Imputation (replaced median)	0.448

4-model ensemble (+ ExtraTrees) and Ensemble weight optimization	0.452
IterativeImputer with LightGBM (replaced KNN)	0.469
Actigraphy features (14 time-series summary stats)	0.470

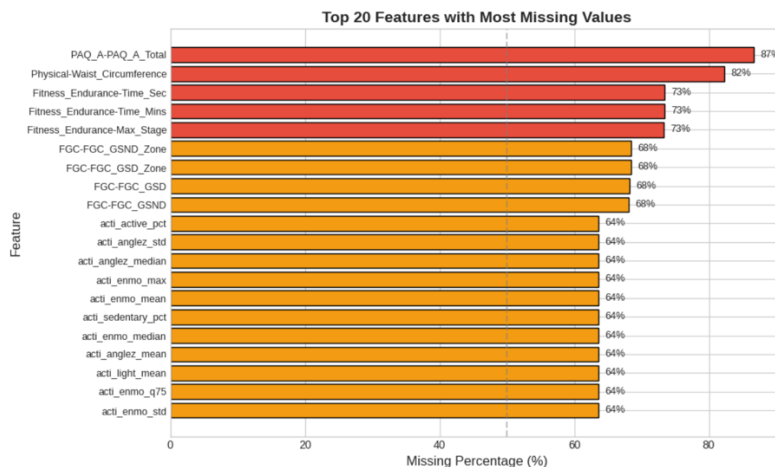
FINAL APPROACH & RESULTS

Our design choices were driven by three main challenges: high missingness, class imbalance, and the coarse, discrete target variable (SII). These challenges guided key decisions throughout the pipeline. Our final pipeline follows a multi-stage approach:

1. Target formulation. Rather than predicting the categorical SII directly (0-3), we train on the continuous PCIAT-Total score (0-100), which allows the model to learn fine-grained differences. The categorical SII labels are recovered at the end via optimized thresholds.



2. Actigraphy features extraction. Only ~25% of participants have accelerometer data. For those who do, we collapse their multi-day time-series into 14 summary statistics capturing movement intensity (ENMO mean, std, median, quartiles, max), activity levels (fraction of time active vs. sedentary), body posture variability (anglez mean, std, median), device wear compliance, ambient light, and recording duration. Participants without actigraphy data receive NaN values for these features, which are handled by the imputer.
3. Imputation. Missing values are filled using an IterativeImputer (MICE-style) with a LightGBM estimator. This approach iteratively predicts each missing feature from all other features using a gradient-boosted tree model, capturing non-linear relationships (e.g., how children's BMI relates to age) that simpler methods like KNN or median imputation cannot.



4. Feature engineering. After imputation, we append basic row-wise aggregate statistics (mean, standard deviation, minimum, maximum across all features per sample) to provide the models with a summary signal of each participant's overall profile.
5. Ensemble of four models. We train four diverse regressors - LightGBM, XGBoost, CatBoost, and ExtraTreesRegressor - within a 10-fold Stratified K-Fold cross-validation (stratified on SII class to maintain class balance across folds). Each model uses early stopping on RMSE to prevent overfitting. Combining diverse models

reduces variance and improves generalization, as different models capture different patterns in the data.

6. Ensemble weight optimization. Rather than equal-weighting the four models, we optimize their blending weights by maximizing Quadratic Weighted Kappa on the out-of-fold predictions using the Nelder-Mead algorithm.
7. Threshold optimization. The blended continuous predictions are converted to SII classes (0-3) using thresholds optimized via Nelder-Mead on the out-of-fold data, starting from the known PCIAT clinical thresholds (30, 49, 79). Since QWK penalizes larger ordinal errors, even small changes in thresholds can significantly impact performance.

Feature importance analysis

To understand which features drive the model's predictions, we averaged LightGBM's split-count feature importances across all 10 cross-validation folds. The most predictive feature by a large margin is SDS_Total_Raw (Sleep Disturbance Scale total score), strongly supporting the hypothesis that sleep quality is a key indicator of problematic internet use. The second most important feature is internet hours per day (PreInt_EduHx-computerinternet_hoursday), which directly measures screen time and serves as a natural predictor.

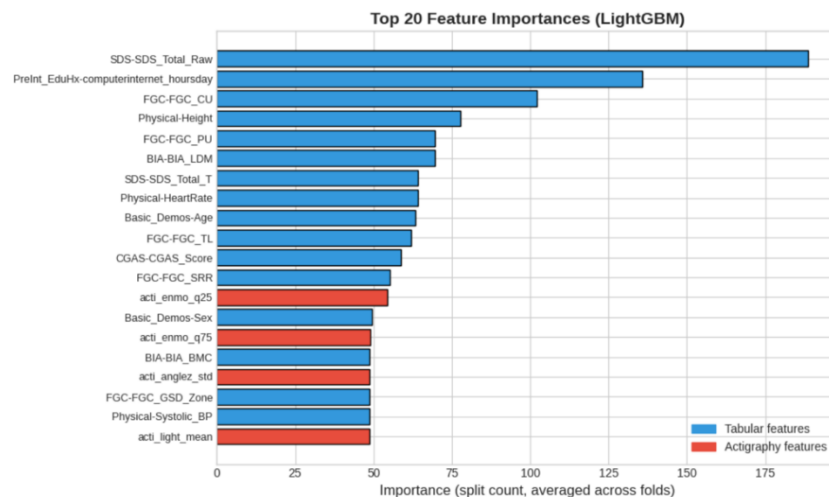
Physical fitness features dominate the middle ranks: FGC_CU (curl-ups), Physical-Height, FGC_PU (push-ups), and FGC_TL/SRR (trunk lift / sit-and-reach) all appear in the top 20, suggesting that children's physical fitness levels are meaningfully associated with internet use severity.

Age and Sex also rank highly, reflecting developmental differences.

Notably, 4 of the top 20 features are actigraphy-derived (shown in red): ENMO quartiles (q25, q75), anglez variability, and ambient light - despite being available for only ~25% of participants. Their presence in the top 20 confirms that accelerometer data carries

meaningful signal about activity patterns that complements the questionnaire-based features.

The figure below visualizes the top 20 most important features described above.



OTHER NOTABLE APPROACHES / ATTEMPTS THAT DIDN'T WORK

- Median imputation (SimpleImputer): Our initial baseline used median imputation for missing values. This was too simple - it ignored relationships between features and hurt model performance. Replacing it with KNN imputation gave a noticeable improvement, and the later switch to IterativeImputer with LightGBM improved it further. KNN imputation assumes locally linear relationships and struggles with the complex, non-linear dependencies in children's health data (e.g., body composition metrics that scale non-linearly with age and sex). The MICE-style approach was a clear upgrade.
- Training directly on SII classes: We initially predicted SII as a categorical target. This discards the ordinal structure and the finer-grained information contained in the continuous PCIAT-Total score. Switching to continuous regression and then converting the predictions to SII using optimized thresholding was a key improvement.
- Single-model approach: Early experiments with a single LightGBM model showed decent results, but the ensemble of four diverse model families (gradient boosting + random forest variant) consistently scored higher through variance reduction.

- Remove implausible physical values (e.g., body fat > 60%) did not improve performance and sometimes reduced it, suggesting the model is stable enough to handle such noise.

LESSONS LEARNED

- Target representation matters. Predicting SII as a 4-class classification problem wastes information, using the underlying continuous PCIAT-Total score as the regression target, then converting back to classes with optimized thresholds, had great impact.
- Imputation is not just a preprocessing step, the choice of imputation method significantly affects performance.
- Ensemble diversity helps. Combining LightGBM, XGBoost, CatBoost, and ExtraTrees - models with fundamentally different splitting strategies - reduced prediction variance more than tuning any single model's hyperparameters.
- Class imbalance significantly impacts performance, especially for higher SII categories, which are underrepresented.

POTENTIAL EXTENSIONS & IMPROVEMENTS

- Domain-specific feature engineering. Create medically meaningful interaction features such as BMI normalized by age group, grip-strength-to-weight ratios, body-fat-to-BMI ratios, and age-sex interaction terms.
- Alternative loss functions.
- Repeated cross-validation. Train across multiple random seeds and aggregate predictions to reduce variance.
- Feature selection. Aggressively prune low-importance or noisy features.